

Kalman filter for attitude determination of a CubeSat using low-cost sensors

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Abstract Nanosatellites have become a common choice for certain space missions, principally in universities, due to their low production and launching costs and reduced developing time. However, they present some technological challenges. In a CubeSat, total mass and available electric power are very restricted. Therefore, subsystem components and payload must be designed to have the smallest size, mass, and power consumption possible for the mission. Regarding to Attitude Determination and Control System, one of restrictions in developing CubeSats is the need to use small and low-cost sensors which fulfill satellite building restrictions. Thus, commercial-off-the-shelf components are a common choice in CubeSat development, but these parts normally present high noise levels. Therefore, a multiplicative Kalman filter for attitude determination algorithm based on quaternions is proposed. Data are simulated using typical values for low-cost sensors, namely, 3-axes magnetometer, 3-axes gyroscope, and a sun sensor. Results are compared to the extended Kalman filter proposed for AAUSAT-3 satellite and to the QUEST method, showing a better precision and low execution time compared to other methods.

Keywords Attitude determination · Kalman filter · Simulation · CubeSat

Mathematics Subject Classification 93E10 · 93E11

1 Introduction

Due to technological advances, smaller satellites are being developed, capable of accomplishing some missions in substitution to large satellites. Although they will never substitute large satellites completely due to limitations in size and power generation, some missions

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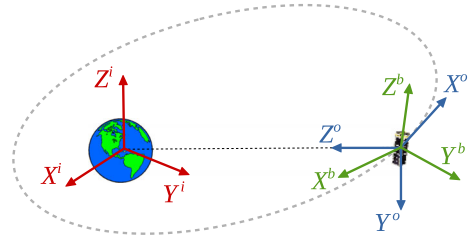
have been designed to operate only with small satellites, such as Earth observation (Selva and Krejci 2012), astrophysical experiments (Ovchinnikov et al. 2014), gather environmental data (Carrara et al. 2014), etc. The small satellites are frequently used for technological demonstrations and education. Small satellites with mass between 1 and 10 kg are classified as nanosatellites (Bouwmeester and Guo 2010).

The recent increase of interest on nanosatellite production was mainly encouraged by CubeSat standard (The CubeSat Program 2015). CubeSat is a building standard for cubic nanosatellites, whose unit has sides of 10 cm and mass of about 1 kg. CubeSats can be built with multiple cubic units. This standardization have eased the access to space by providing low launch costs, once CubeSats can be carried as secondary payloads on launch vehicle. Although initiated in universities as engineering teaching instrument, CubeSat applications have expanded to other sectors, as private companies and governmental organizations (Swartwout 2013).

Nanosatellites have become a common choice for certain space missions, principally in universities, due to their low production and launching costs and reduced developing time. However, they present some technological challenges. In a CubeSat, total mass and available electric power are very restricted. Therefore, subsystem components and payload must be designed to have the smallest size, mass, and power consumption possible for the mission. Regarding the attitude determination and control system (ADCS), a problem in its development for CubeSats is the need to use small, low-cost sensors and actuators which meet the constraints of satellite design. Therefore, magnetic coils as attitude control actuator is a common choice for CubeSats. To reduce development and production costs, CubeSats are commonly built with commercial-off-the-shelf (COTS) components, which tend to present high noise levels. Therefore, attitude determination methods should be designed to minimize the impact of high noises.

In this work, a multiplicative Kalman filter algorithm for attitude determination, based on quaternions, is proposed. Quaternions are overdetermined in attitude representation, leading to a singularity in covariance matrix. The Kalman filter is carried out with a reduced state and the quaternion is recovered such that unit norm is preserved. The proposed filter uses 3 axes magnetometer, sun sensor, and 3 axes gyroscope data, simulated for a 3U CubeSat in a low Earth orbit. Results are compared to the extended Kalman filter proposed in Jensen and Vinther (2010) for AAUSAT-3 attitude determination and to the QUEST method (Shuster and Oh 1981).

Several papers have been published in hardware and software development for attitude determination and control. Jensen and Vinther (2010) describes the design and analysis of an ADCS for AAUSAT-3 mission. A quaternion implementation of Extended and Unscented Kalman Filters, using a low cost of the shelf sensor setup, is proposed. Theil et al. (2003) presents a low cost, robust Attitude Determination System with good accuracy for small satellites, achieved using a combination of magnetometer and coarse sun sensor. Earth's albedo error is considered on simulation of the coarse sun sensor measurement. Duarte et al. (2009, 2011) address the practical implementation and the performance analysis of a fault tolerant attitude determination procedure implemented in a microprocessor environment, able to properly calculate the attitude using an architecture based on low-cost COTS components. Carrara et al. (2014) describes the ADCS operation modes of CONASAT mission, an 8U CubeSat which is being designed to gather environmental data. The results from the simulated attitude determination, estimation, and control are addressed, focusing attitude performance for both nominal and safe modes. Garcia et al. (2012, 2016a, b) use real data from CBERS-2 sensors to analyze an unscented Kalman filter in attitude estimation problems and compare it with the standard estimator for non-linear estimation problems, using reduced state.

Fig. 1 Coordinate systems

2 Satellite model

This section describes coordinate systems and mathematical models of satellite attitude kinematics and dynamics.

2.1 Coordinate systems

The inertial system has its origin at Earth center, Z^i axis is pointed to north pole and coincident to Earth rotation axis, X^i axis points to Vernal Equinox, and Y^i axis is perpendicular to other axis, completing the right-handed frame.

The orbital system is centered on satellite center of mass, its Z^o axis points to nadir direction, X^o axis is parallel to satellite velocity vector, and Y^o axis is anti-parallel to orbital angular moment vector.

The body system is also centered on satellite center of mass, and its axes (X^b , Y^b , Z^b) lie along the directions of principal moments of inertia. These systems are shown in Fig. 1.

The satellite is modeled as an Earth pointing mission, so its attitude is described as orientation of body frame related to orbital frame.

2.2 Dynamic and kinematic models

The dynamic model is represented by Euler's moment equation, given by [Crassidis and Junkins \(2012\)](#) and [Jensen and Vinther \(2010\)](#):

$$\dot{\omega}_{ib}^b = \mathbf{J}^{-1} \left[\mathbf{N}^b - \mathbf{S}(\omega_{ib}^b) \mathbf{J} \omega_{ib}^b \right] \quad (1)$$

where the vector $\omega_{ib}^b = [\omega_1 \ \omega_2 \ \omega_3]^T$ represents angular velocity of body frame with respect to orbital frame, $\mathbf{J}$ is inertia matrix, $\mathbf{S}(\omega_{ib}^b)$ is a skew-symmetric matrix which performs a vectorial product:

$$\mathbf{S}(\omega_{ib}^b) = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \quad (2)$$

and $\mathbf{N}^b$ represents the total torque actuating on satellite in body frame, which includes perturbation torques as gravity gradient, and control torque generated by magnetic coils. The subscripts ib means body related to inertial frames, and superscript b means written in body frame.

The satellite attitude is represented by a quaternion, which represents orientation of body frame related to orbital frame:

$$\mathbf{q}_o^b = [q_1 \ q_2 \ q_3 \ q_4]^T \quad (3)$$

where q_4 is the scalar part of quaternion.

The quaternion kinematics is given by [Crassidis and Junkins \(2012\)](#) and [Jensen and Vinther \(2010\)](#):

$$\dot{\mathbf{q}}_o^b = \frac{1}{2} \boldsymbol{\Omega} \left(\boldsymbol{\omega}_{ob}^b \right) \mathbf{q}_o^b \quad (4)$$

where

$$\boldsymbol{\Omega} \left(\boldsymbol{\omega}_{ob}^b \right) = \begin{bmatrix} -\mathbf{S} \left(\boldsymbol{\omega}_{ob}^b \right) & \boldsymbol{\omega}_{ob}^b \\ -\left(\boldsymbol{\omega}_{ob}^b \right)^T & 0. \end{bmatrix} \quad (5)$$

The angular velocities $\boldsymbol{\omega}_{ib}^b$ and $\boldsymbol{\omega}_{ob}^b$ are related:

$$\begin{aligned} \boldsymbol{\omega}_{ib}^b &= \boldsymbol{\omega}_{ob}^b + \boldsymbol{\omega}_{io}^b \\ &= \boldsymbol{\omega}_{ob}^b + \mathbf{A} \left(\mathbf{q}_o^b \right) \boldsymbol{\omega}_{io}^o \end{aligned} \quad (6)$$

where $\mathbf{A} \left(\mathbf{q}_o^b \right)$ the transformation matrix from orbital to body system, and $\boldsymbol{\omega}_{io}^o = [0 \ -\omega_o \ 0]^T$ is the orbital angular velocity vector.

3 Kalman filter

The Multiplicative Extended Kalman Filter (MEKF) was originally introduced in [Lefferts et al. \(1982\)](#), to deal with quaternion non-independent components. In describing attitude, the four elements of quaternion are constrained to unit norm, leading to a singular covariance matrix in additive measurement update Kalman filter.

The MEKF introduced a multiplicative error state $\delta \mathbf{q}$, which represents a small rotation from the predicted to the expected attitude:

$$\delta \mathbf{q} = \mathbf{q} \otimes \hat{\mathbf{q}}^{-1}. \quad (7)$$

The symbol $\otimes$ means quaternion multiplication operation. The multiplicative state overcomes the problem of maintaining unit quaternion length, and the reduced representation avoids a singular covariance matrix.

The model for the gyroscope measurement is given by [Lefferts et al. \(1982\)](#):

$$\boldsymbol{\omega} = \boldsymbol{\omega}_m - \boldsymbol{\beta} - \boldsymbol{\eta}_\omega \quad (8)$$

$$\dot{\boldsymbol{\beta}} = \boldsymbol{\eta}_\beta \quad (9)$$

where $\boldsymbol{\eta}_\omega$ and $\boldsymbol{\eta}_\beta$ are zero mean Gaussian white noise processes with covariances given by $\sigma_\omega^2 \mathbf{I}_{3 \times 3}$ and $\sigma_\beta^2 \mathbf{I}_{3 \times 3}$, respectively, $\boldsymbol{\beta}$ is a bias vector, and $\boldsymbol{\omega}_m$ is the measured observation. $\mathbf{I}_{3 \times 3}$ represents identity matrix. The angular velocity can be estimated by [Lefferts et al. \(1982\)](#)

$$\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}_m - \hat{\boldsymbol{\beta}}. \quad (10)$$

Here, the circumflex ($\hat{\cdot}$) means estimated value for a variable.

Therefore, from Eqs. (4) and (9), the state space model for the filter can be written as

$$\begin{aligned} \dot{\mathbf{q}} &= \frac{1}{2} \boldsymbol{\Omega} \left(\hat{\boldsymbol{\omega}} \right) \mathbf{q} \\ \dot{\boldsymbol{\beta}} &= 0. \end{aligned} \quad (11)$$

The measurements from a sun sensor and a magnetometer are used to update quaternions and gyroscope bias in the filter. The measurements are taken in body frame, while the expected measurements are calculated in inertial frame. This relation is given by

$$\mathbf{r}^b = \mathbf{A} \left(\mathbf{q}_i^b \right) \mathbf{r}^i \quad (12)$$

where $\mathbf{r}^i$ is the measurement vector expressed in inertial frame (either sun vector or magnetic field), $\mathbf{r}^b$ is the measurement vector in body frame, and $\mathbf{A} \left(\mathbf{q}_i^b \right)$ represents the transformation matrix from inertial to body frames.

Therefore, the estimated measurement $\hat{\mathbf{r}}^b$ is given by

$$\hat{\mathbf{r}}^b = \mathbf{A} \left(\hat{\mathbf{q}}_i^b \right) \hat{\mathbf{r}}^i \quad (13)$$

and can be related to quaternion increment $\delta \mathbf{q}$ (Crassidis and Junkins 2012):

$$\delta \mathbf{r} = 2\mathbf{S} \left(\mathbf{A} \left(\hat{\mathbf{q}}_i^b \right) \mathbf{r}^i \right) \delta \mathbf{q}. \quad (14)$$

State space and measurement models can be written in a discrete form to implement the Kalman filter time and measurement update steps.

3.1 Time update

To perform a discrete estimation, the model equation (11) can be written in discrete form (Crassidis and Junkins 2012):

$$\hat{\mathbf{q}}_{k+1}^- = \Theta \left(\hat{\mathbf{q}}_k^+ \right) \hat{\mathbf{q}}_k^+ \quad (15)$$

$$\hat{\beta}_{k+1}^- = \hat{\beta}_k^+ \quad (16)$$

with $\hat{\omega}_k^+ = \omega_m - \hat{\beta}_k^+$ and

$$\Theta \left(\hat{\omega}_k^+ \right) = \begin{bmatrix} \cos \left(\frac{1}{2} \|\hat{\omega}_k^+\| \Delta t \right) \mathbf{I}_{3 \times 3} - \mathbf{S} \left(\hat{\psi}_k^+ \right) & \hat{\psi}_k^+ \\ - \left(\hat{\psi}_k^+ \right)^T & \cos \left(\frac{1}{2} \|\hat{\omega}_k^+\| \Delta t \right) \end{bmatrix} \quad (17)$$

$$\hat{\psi}_k^+ = \frac{\sin \left(\frac{1}{2} \|\hat{\omega}_k^+\| \Delta t \right) \hat{\omega}_k^+}{\|\hat{\omega}_k^+\|}$$

where $\|\cdot\|$ represents vector norm, minus sign ($-$) means propagated state before measurement update and plus sign ($+$) means state after measurement update, and Δt is the update interval.

The covariance matrix can be updated using a state transition matrix Φ_k :

$$\mathbf{P}_{k+1}^- = \Phi_k \mathbf{P}_k^+ \Phi_k^T + \mathbf{G}_k \mathbf{Q}_k \mathbf{G}_k^T \quad (18)$$

where

$$\Phi_k = \begin{bmatrix} \Phi_{11} & \Phi_{12} \\ \Phi_{21} & \Phi_{22} \end{bmatrix} \quad (19)$$

and

$$\begin{aligned} \Phi_{11} &= \mathbf{I}_{3 \times 3} - \mathbf{S} \left(\hat{\omega}_k^+ \right) \frac{\sin \left(\|\hat{\omega}_k^+\| \Delta t \right)}{\|\hat{\omega}_k^+\|} + \mathbf{S} \left(\hat{\omega}_k^+ \right)^2 \frac{[1 - \cos \left(\|\hat{\omega}_k^+\| \Delta t \right)]}{\|\hat{\omega}_k^+\|^2} \\ \Phi_{12} &= \mathbf{S} \left(\hat{\omega}_k^+ \right) \frac{[1 - \cos \left(\|\hat{\omega}_k^+\| \Delta t \right)]}{\|\hat{\omega}_k^+\|^2} - \mathbf{I}_{3 \times 3} \Delta t \\ &\quad - \mathbf{S} \left(\hat{\omega}_k^+ \right)^2 \frac{[\|\hat{\omega}_k^+\| \Delta t - \sin \left(\|\hat{\omega}_k^+\| \Delta t \right)]}{\|\hat{\omega}_k^+\|^3} \end{aligned} \quad (20)$$

$$\Phi_{21} = \mathbf{0}_{3 \times 3}$$

$$\Phi_{22} = \mathbf{I}_{3 \times 3}$$

Using the discrete form, the covariance of the process noise matrix $\mathbf{Q}_k$ and $\mathbf{G}_k$ is given by

$$\mathbf{Q}_k = \begin{bmatrix} \left(\sigma_\omega^2 \Delta t + \frac{1}{3} \sigma_\beta^2 \Delta t^3 \right) \mathbf{I}_{3 \times 3} - \left(\frac{1}{2} \sigma_\beta^2 \Delta t^2 \right) \mathbf{I}_{3 \times 3} \\ - \left(\frac{1}{2} \sigma_\beta^2 \Delta t^2 \right) \mathbf{I}_{3 \times 3} \quad \left(\sigma_\beta^2 \Delta t \right) \mathbf{I}_{3 \times 3} \end{bmatrix} \quad (21)$$

$$\mathbf{G}_k = \begin{bmatrix} -\mathbf{I}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_{3 \times 3} \end{bmatrix} \quad (22)$$

3.2 Measurement update

The quaternion measurement update is calculated in multiplicative form, considering the error quaternion in Eq. (7). The filter updates the reduced error quaternion $\delta \mathbf{q} = [\delta q_1 \ \delta q_2 \ \delta q_3]^T$, and the fourth element δq_4 is calculated to keep the quaternion unity norm. Then, the updated quaternion is recovered from

$$\delta \mathbf{q} = \mathbf{q}_k^+ \otimes (\mathbf{q}_k^-)^{-1}. \quad (23)$$

Therefore, filter measurement update equations are [Lefferts et al. \(1982\)](#) and [Jensen and Vinther \(2010\)](#):

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}_k^T \left(\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k \right)^{-1} \\ \mathbf{P}_k^+ &= (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- \\ \begin{bmatrix} \delta \mathbf{q}_k \\ \delta \boldsymbol{\beta}_k \end{bmatrix} &= \mathbf{K}_k \left(\begin{bmatrix} \mathbf{m}_k^b \\ \mathbf{s}_k^b \end{bmatrix} - \begin{bmatrix} \hat{\mathbf{m}}_k^b \\ \hat{\mathbf{s}}_k^b \end{bmatrix} \right) \\ \mathbf{q}_k^+ &= \left[\delta \mathbf{q}_k^T \sqrt{1 - \delta \mathbf{q}_k^T \cdot \delta \mathbf{q}_k} \right]^T \otimes \mathbf{q}_k^- \\ \boldsymbol{\beta}_k^+ &= \boldsymbol{\beta}_k^- + \delta \boldsymbol{\beta}_k. \end{aligned} \quad (24)$$

$\mathbf{H}_k$ matrix arises from measurement linearization and $\mathbf{R}_k$ is the measurement covariance matrix:

$$\mathbf{H}_k = \begin{bmatrix} 2\mathbf{S}(\mathbf{m}^i) & \mathbf{0}_{3 \times 3} \\ 2\mathbf{S}(\mathbf{s}^i) & \mathbf{0}_{3 \times 3} \end{bmatrix}, \quad \mathbf{R}_k = \begin{bmatrix} \sigma_m^2 \cdot \mathbf{I}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \sigma_s^2 \mathbf{I}_{3 \times 3} \end{bmatrix} \quad (25)$$

where $\mathbf{m}^b$ and $\mathbf{s}^b$ represent magnetometer and sun sensor measurements, and σ_m^2 and σ_s^2 their variances, respectively.

3.3 Detumbling controller

Detumbling controller is used to reduce the satellite angular velocity to values, such that other control mode can take place. Here, a B-dot control is introduced in simulation to slow down the high initial angular velocity. The B-dot control law is given by the equation ([Gravdahl 2004](#)):

$$\mathbf{M}^b = -K \dot{\mathbf{B}}^b + \mathbf{M}_c \quad (26)$$

where $\mathbf{M}^b$ is magnetic moment which coils must apply, $\dot{\mathbf{B}}^b$ is the magnetic field change rate, $\mathbf{M}_c$ is a internal residual magnetic moment, and K is the controller gain. The generated control torque $\mathbf{N}_c^b$ is

$$\mathbf{N}_c^b = \mathbf{M}^b \times \mathbf{B}^b. \quad (27)$$

All quantities are written in the body frame.

4 Simulation and results

Simulated satellite was defined as 3U CubeSat with 10 cm along X^b and Y^b sides, and 30 cm along Z^b side, and total mass of 3.3 kg. This configuration results in a diagonal inertia matrix: $\mathbf{J} = \text{diag}[0.0275 \ 0.0275 \ 0.0055] \text{ kg m}^2$. Data were obtained for a circular orbit with altitude of 500 km and inclination of 45° . The simulation was carried out in Matlab language.

To simulate measurement outputs from magnetometer and sun sensor, wide band noise was added to values given by Earth magnetic field and sun position models. The Earth magnetic field was calculated using IGRF-12 model (Thébault et al. 2015) and wide band noise in magnetometer has the standard deviation of $3 \times 10^{-7} \text{ T}$. The sun vector measurement is calculated adding a wide band noise at azimuth and elevation angles of modeled sun position in satellite body, with the standard deviation of 0.035 rad. The sun position model is described in Reda and Andreas (2008). It was considered that sun vector can be obtained whenever the sun is hitting satellite surfaces. The simulated gyroscope output $\tilde{\omega}$ consists of the true vehicle rotation rate ω (given by models in simulation) plus a constant offset b , a moving bias c , and wide band sensor noise η (Flenniken 2005), as shown in Eq. 28:

$$\tilde{\omega} = \omega + b + c + \eta, \quad \begin{cases} b = [0.035 \ -0.017 \ 0.009]^T \text{ rad/s} \\ \dot{c} = v, \quad v \sim N(0; 2 \times 10^{-4} \text{ rad/s}^2) \\ \eta \sim N(0; 3 \times 10^{-2} \text{ rad/s}) \end{cases} \quad (28)$$

The notation $N(\mu; \sigma)$ means normal distribution with mean μ and standard deviation σ . The noise values in simulated sensor models correspond to typical low-cost sensors ones.

Sensor data were sampled at 1 Hz and attitude determination algorithms calculate their estimates at the same rate. Detumbling control was kept on during whole simulation, with controller gain set to $K = 7000$. However, during a sampling time, it was simulated a period of 100 ms when coils are turned off to allow magnetometers measure Earth magnetic field with no interference. Then, coils are turned on for the remaining sample time, or 900 ms.

Satellite initial angular velocity was set to $\omega_{ob}^b = [-0.16 \ 0.07 \ -0.11]^T \text{ rad/s}$. Filter initialization parameters for the proposed and AAUSAT-3 filters were adjusted to

- Initial states: $\mathbf{q}_0 = [0 \ 0 \ 0 \ 1]^T$; $\boldsymbol{\beta}_0 = [0 \ 0 \ 0]^T \text{ rad/s}$
- Initial state covariance:
 $\mathbf{P}_0 = \text{diag}([0.5^2 \ 0.5^2 \ 0.5^2 \ (0.1 \text{ rad/s})^2 \ (0.1 \text{ rad/s})^2 \ (0.1 \text{ rad/s})^2])$
- Measurements covariance: $\sigma_m = 1 \times 10^{-6} \text{ T}$; $\sigma_s = 0.04 \text{ rad}$
- Process covariance: $\sigma_\omega = 4.89 \times 10^{-3} \text{ rad/s}^{1/2}$, and $\sigma_\beta = 3.14 \times 10^{-4} \text{ rad/s}^{3/2}$ in $\mathbf{Q}_k$ matrix (Eq. 21)

Figure 2 shows the evolution of Euler angles (first column) and angular velocity components (second column) during five orbits. Figure 3 shows gyroscope simulated bias (dark curve) and proposed filter estimated bias (white curve inside dark one). Covariance of reduced state is shown in Fig. 4. In this figure, the first column brings square root values of diagonal elements corresponding to quaternion increment, and the second column, square root values of diagonal elements corresponding to bias. AAUSAT-3 filter has similar bias and covariance curves, so they are not presented.

The epochs when satellite is in eclipse can be noticed in Fig. 4. During the eclipse, the standard deviation curve has higher values, leading to unreliable estimates from the Kalman filter. This occurs, e.g., between about 0.5 and 0.9 orbit.

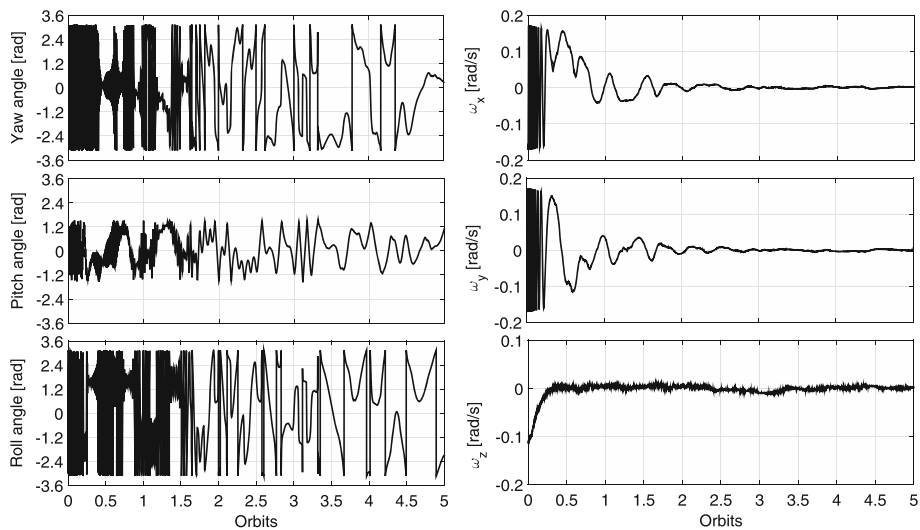


Fig. 2 Simulated Euler angles and angular velocity

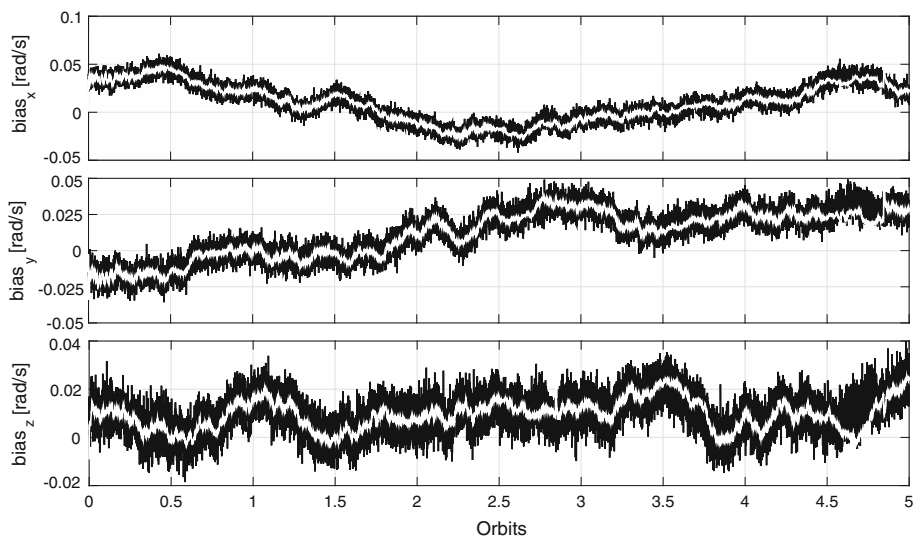


Fig. 3 Simulated and estimated bias. The *dark curve* represents simulated bias, and the *white curve* represents estimated bias

The precision of attitude determination methods is compared through Euler angle errors for each method. The Euler angle error is the difference between estimated and “true” angles. The former is calculated from estimated quaternion and the latter from simulated data. Figure 5 shows the Euler angle error for the proposed filter, Fig. 6 shows the Euler angle error for the AAUSAT-3 filter, and Fig. 7 shows the Euler angle error using QUEST method. Mean and standard deviation statistics for each component are shown inside the respective graph, calculated excluding eclipse epochs.

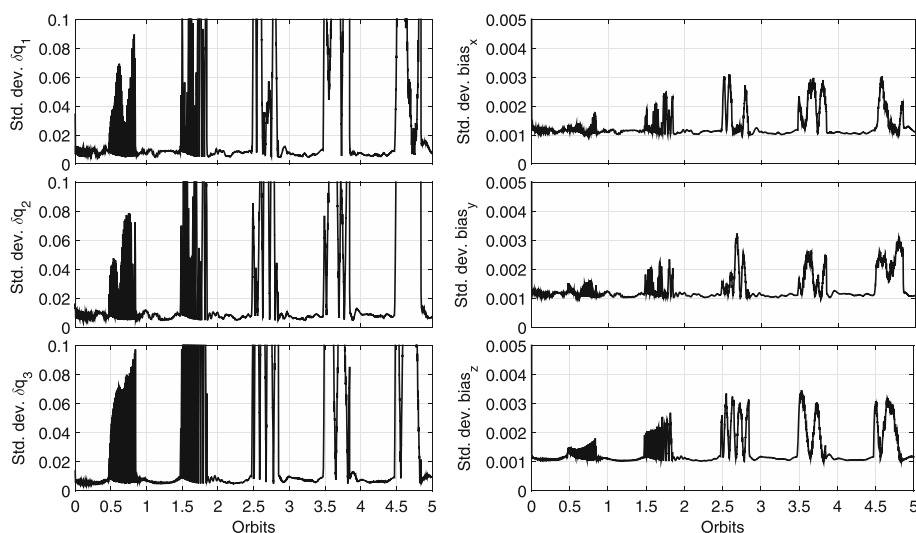


Fig. 4 Covariances of estimated states

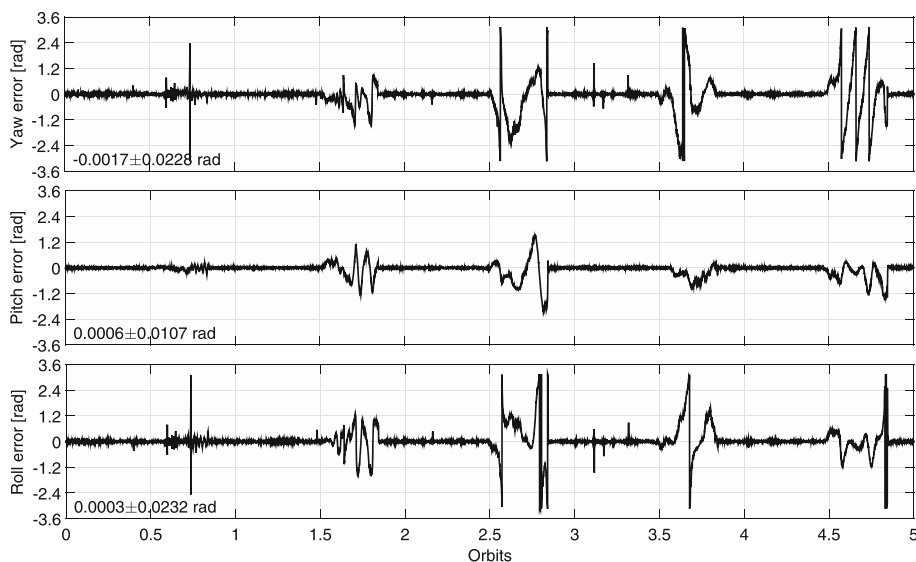


Fig. 5 Euler angles error from the proposed filter

Figure 8 presents execution time for each attitude determination algorithm, in addition to their mean and standard deviation statistics. These values were obtained in a Core i5-3570, 8 GB de RAM computer running on Windows 7. QUEST method has the lowest execution time, because it is an optimal deterministic method. However, its precision is also the lowest, about 0.06 rad. Execution time of AAUSAT-3 filter is about 20 times longer than QUEST filter, and achieved a precision of 0.03 rad. This filter uses a numerical integrator in time update step, leading to a longer execution time. The proposed filter has an execution time only 3 times longer than QUEST method, and precision of 0.02 rad, comparable with AAUSAT-3 filter.

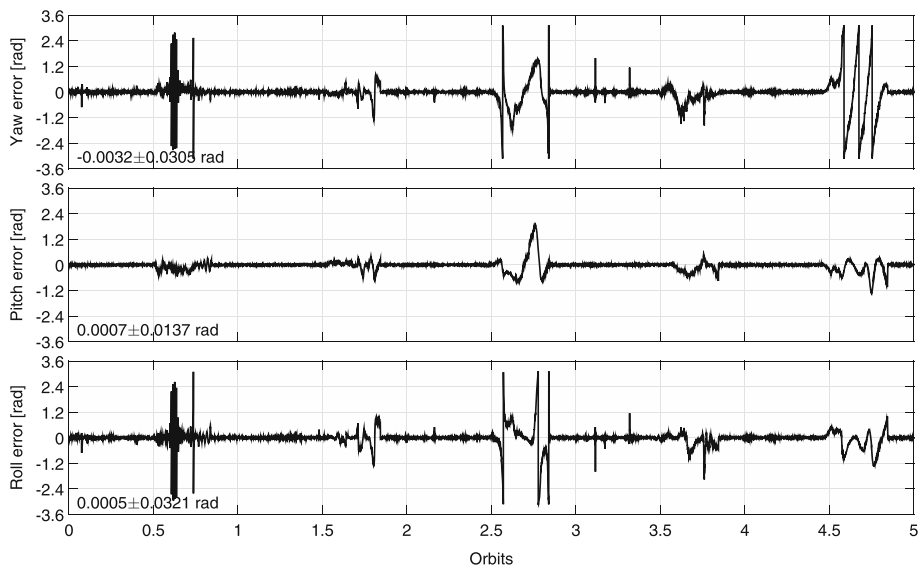


Fig. 6 Euler angles error from AAUSAT-3 filter

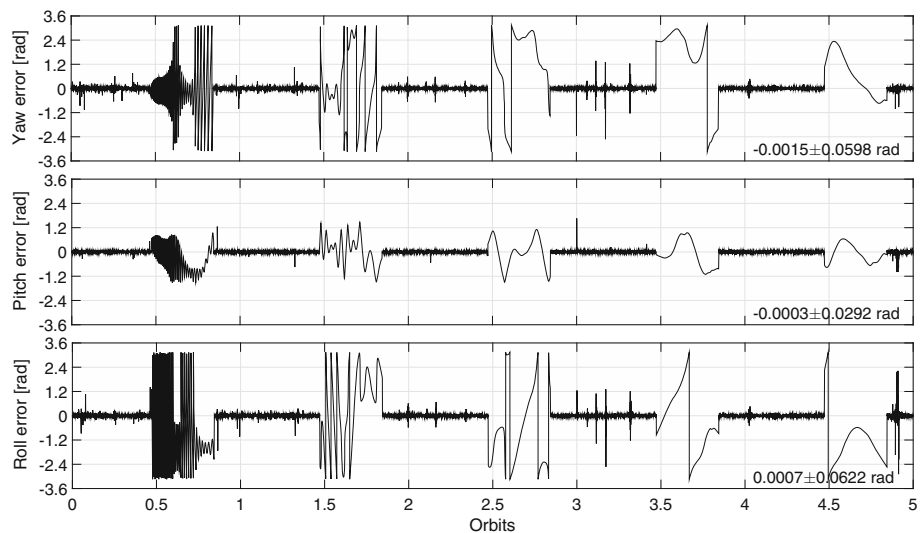


Fig. 7 Euler angles error from QUEST

5 Conclusion

A multiplicative Kalman filter for CubeSat attitude determination was implemented. Sensor data, namely magnetometer, sun sensor, and gyroscope, were obtained from simulation of a CubeSat placed in low Earth orbit. States estimated by the filter were attitude quaternions and gyroscope bias, with the latter being used to calculate satellite angular velocity. These estimates were compared to values from other two attitude determination algorithms: filter proposed to AAUSAT-3 satellite and QUEST method.

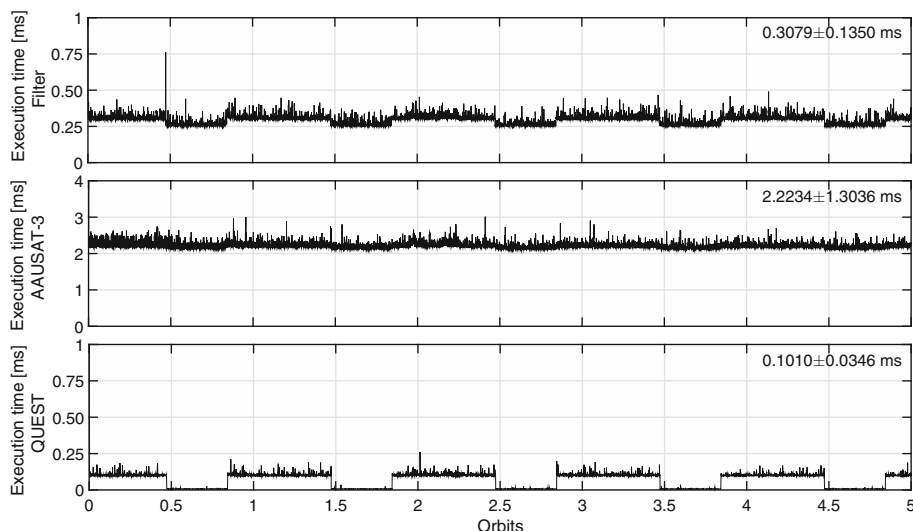


Fig. 8 Execution times

From filter results, Euler angles calculated from estimated quaternions had precision of 0.02 rad ($\approx 1.2^\circ$). These values are similar to AAUSAT-3 ones. Euler angles from QUEST showed precision of 0.06 rad ($\approx 3.4^\circ$). Although all algorithms were kept running during whole simulation, the lack of sun sensor data during eclipses makes the estimates unreliable. Therefore, a strategy for attitude estimation during eclipses should be developed.

The execution time of the proposed filter was only three times longer than QUEST method, which is based on optimization technique. AAUSAT-3 filter had a much longer execution time, because its time update phase is carried out by continuous propagation.

Although these results can show this filter feasibility on simulated data, tests should be carried out using different data sets of initial conditions, as well as embedded software, and in hardware to confirm these results and its integration on an attitude control system.

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